Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/05/2025
PI Name:	Wozniak, Katherine Ph.D.		
Project Title:	Defining Gut Microbial Changes to Space-like Radiation to Develop a Radiation-Resistant Microbiome		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030-3411	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2023 TRISH-RFA-2301: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	08/01/2023	**End Date:**	07/31/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Britton, Robert (MENTOR: Baylor College Of Medicine, Inc.) Blutt, Sarah (Baylor College Of Medicine, Inc.)		
Grant/Contract No.:	NNX16AO69A-P0801		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Long-term missions to Mars will have detrimental effects on the human body. The human gastrointestinal tract, referred to as the human gut, completely regenerates every 3-7 days and is therefore a target of space radiation. Within the human gut, millions of microorganisms aid in digestion and absorption of nutrients and have been linked to overall health. It has been shown that space travel changes the human gut microbiome, likely due in part to space radiation. The detrimental effects of space radiation will be more severe with increased time in space and therefore merit strategies to maintain astronaut health. This proposal seeks to understand how gut microbiome members change during space-like radiation and how these changes adversely affect human health. Using spacelike radiation, we will irradiate gut microbial communities and uncover changes in membership using metagenomics. Irradiated communities will be administered to intestinal organoids to measure the functional effect on organoids, including proliferation, barrier function, and nutrient absorption. In addition, we aim to create a radiation-resistant gut microbial community using selection and bioengineering that will help mitigate detrimental effects of space radiation from within the body. Together, this study will help us understand how human gut microbiome structure and function changes in space and will develop a curated microbial community to protect against radiation damage in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Aim 1: This year with streamlined protocols for fecal microbial community stabilization in minibioreactor arrays (MBRAs) and irradiation, I was able to conduct 4 bioreactor runs (2 with 2 communities and 1 dose of irradiation and 2 with 1 community and 3 doses of irradiation). With multiple repeated experiments spanning up to 8 biological replicates in each group, we are now able to assess radiation-induced microbiome changes. Aim 2: I continued honing my human intestinal organoid (HIOs) experimentation skills and streamlined my methodology to differentiate HIOs into villus-like cells. I performed high dose X-ray radiation on 4 different jejunal lines for RNA-sequencing as a proof-of-concept and described their general response. I also measured IFN-lambda levels from differentiated HIOs overlaid with microbial community supernatants and am continuing this line of investigation now that I see concrete community abundance changes. I am currently testing the ability of Limosilactobacillus reuteri-secreted growth factors to stimulate proliferation of villus-like HIOs. So far, we can see that L. reuteri supernatants containing engineered growth factors promote proliferation in differentiated enteroids greater than medium controls. In the coming year (unfunded), I will select community samples for RNA-sequencing to determine what the microbial community response to radiation is in both highly perturbed communities and unperturbed communities. I am expanding the L. reuteri engineered factors work to test different HIO lines assess the supernatant ability to stimulate proliferation following radiation.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Articles in Peer-reviewed Journals	Lakey BD, Wozniak KJ, Britton RA, Tabor JJ. "Mucin-derived sugars act as metabolic brakes controlling growth initiation in Akkermansia muciniphila." Gut Microbes. 2026 Jun 26;18(1):2691334. https://doi.org/10.1080/19490976.2026.2691334 ; PMID: 42359789; PMCID: PMC13313206 , Jun-2026
Awards	Wozniak, KJ "Cohort-5 participant, NASA Spaceflight Technology, Applications, and Research (STAR) virtual course" Sep-2024
Awards	Wozniak, KJ "NIH T32 Awardee, Antimicrobial Resistance Training Program, Texas Medical Center" Sep-2025
Awards	Britton R "Center for Space Medicine Space Health Pilot Award (cancelled due to budget cuts)" Jul-2025
Awards	Britton B, Blutt S "Engineered Microbes to Mitigate Intestinal Radiation Damage (NIH R21)" Jul-2025